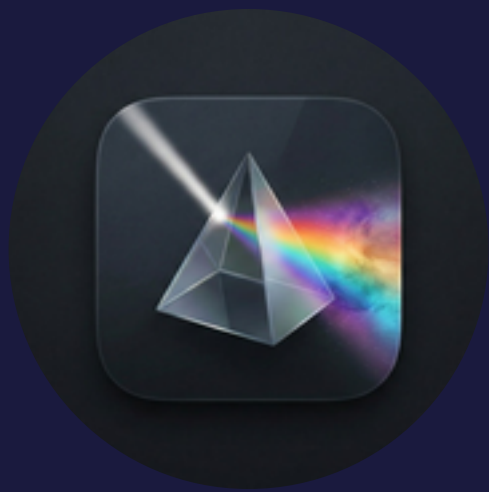




Unfiltered

Your Emotional Companion

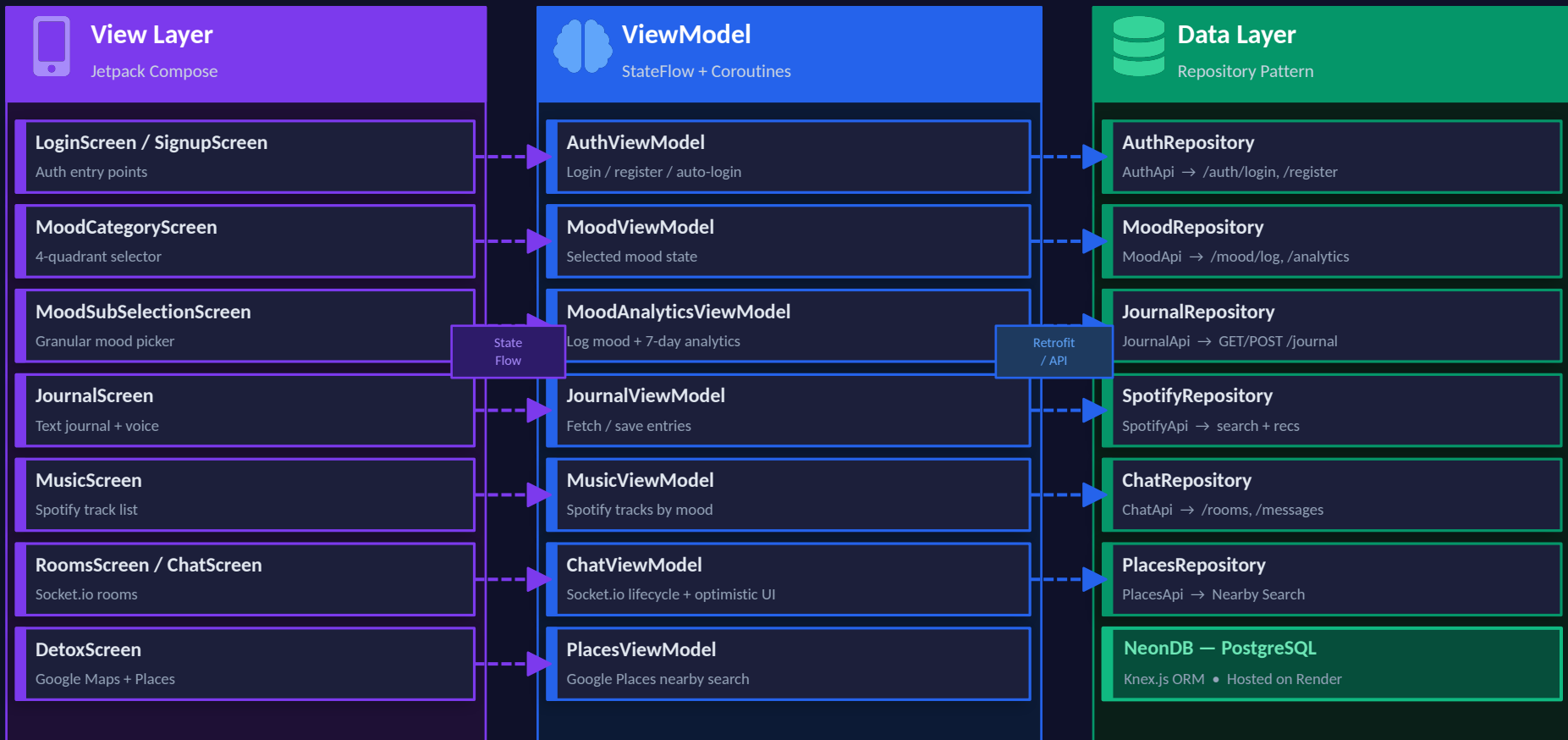
Update 2

CS 501 E1: Mobile Application Development

Kaushik Sandeep Joshi • Ashish Sandeep Joshi • Vidney Milind Jadhav

MVVM Architecture

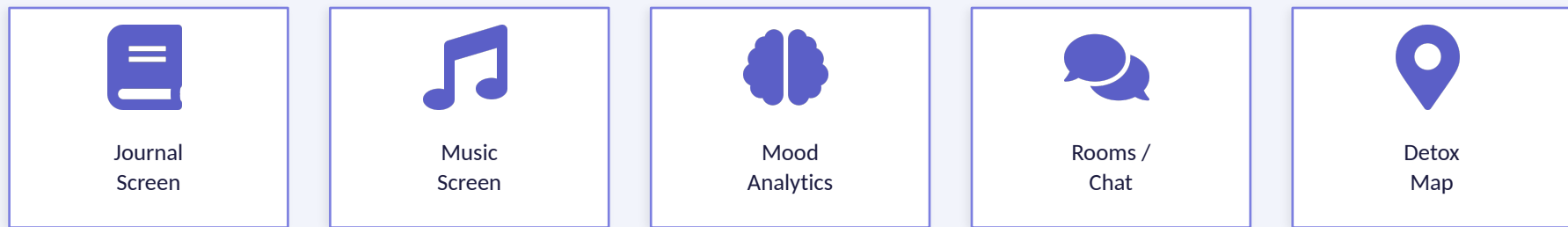
Unfiltered App — Kotlin + Jetpack Compose



Navigation Flow & App Screens



NavHost routes to feature screens:



Bottom `NavigationBar` provides persistent access • Arguments passed via typed `NavHost` routes (e.g. `MoodCategory` → `MoodSubSelection(modeType)`)

Database Schema & Backend Deployment

NeonDB

Render

users

id serial PK
email varchar(255)
password_hash varchar
name varchar(255)
created_at timestamp

daily_moods / mood_logs

id serial PK
user_id FK → users
date date
mood_value varchar
quadrant varchar

journals / journal_entries

id serial PK
user_id FK → users
text / content
mood_tag varchar
created_at timestamp

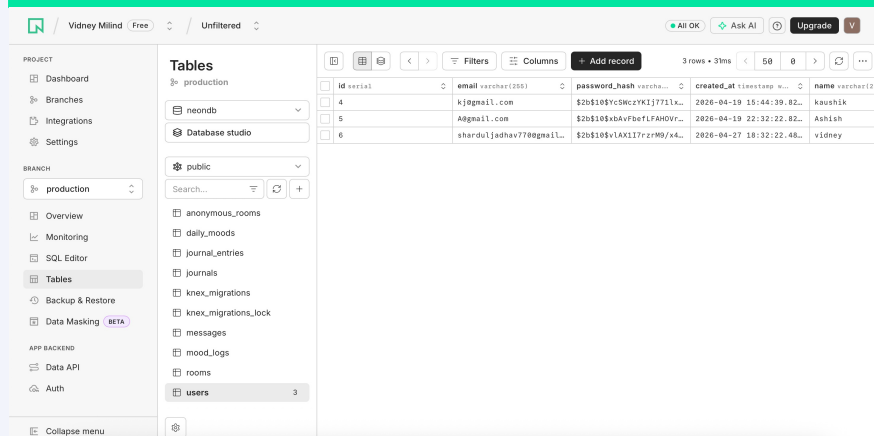
rooms + messages

room_id serial PK
name / mood_tag varchar
message_id serial PK
room_id FK → rooms
content + sender + timestamp

knex_migrations

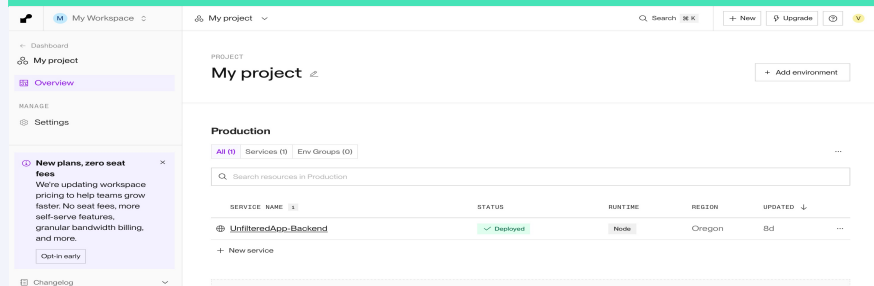
Knex migration tracking

NeonDB — Production Database (PostgreSQL)



id serial	email varchar(255)	password_hash varchar(255)	created_at timestamp w/ time zone	name varchar(255)
4	kj@gmail.com	\$2b\$10\$Yc5WczYK1j771L...	2026-04-19 15:44:39.8Z.	kaushik
5	A@gmail.com	\$2b\$10\$b6VbPbfLFAHOVr...	2026-04-19 22:32:22.8Z.	Ashish
6	sharduljadhav77@gmail.com	\$2b\$10\$V1AX17rzz99/x4...	2026-04-27 10:32:22.48...	vidney

Render — Backend Deployment (Node.js · Oregon · Deployed)



SERVICE NAME	STATUS	RUNTIME	REGION	UPDATED
UnfilteredApp:Backend	✓ Deployed	Node	Oregon	8d

PostgreSQL hosted on NeonDB • Node.js/Express backend deployed on Render (Oregon) • ORM: Knex.js • Auth: JWT tokens

Debugging & Testing Strategy



Runtime Debugging

- Android Studio Logcat for ViewModel & StateFlow logs
- Breakpoints in ViewModels to inspect state transitions
- Network Inspector for API call tracing (Retrofit)
- Socket.io event logging for real-time chat debugging



Manual Feature Testing

- End-to-end flow: Login → Mood → Journal → Music → Rooms
- Mood quadrant navigation verified across all 4 states
- Journal text & voice entry tested on physical device
- Maps & nearby places tested with live GPS location



Auth & State Testing

- JWT token persistence tested across app restarts
- Auto-login flow (SplashScreen → auto-auth check)
- Session expiry and logout state clearing validated
- SharedPreferences token read/write verified



API & Data Testing (NeonDB)

- Postman used to validate all Express API endpoints
- NeonDB Database Studio used to inspect live table data
- Knex migrations verified on production NeonDB schema
- Spotify Client Credentials token refresh tested
- Render deployment logs checked for runtime errors

Team Workflow Breakdown

Kaushik Sandeep Joshi

Backend & API

- Built Express.js backend with Auth APIs (login/register/me)
- Designed PostgreSQL schema using Knex migrations on NeonDB
- Configured API routes: auth, mood, journal, rooms
- Deployed backend to Render (Oregon, Node.js service)
- Postman testing of all API endpoints

Ashish Sandeep Joshi

UI/UX Development

- Built all Jetpack Compose screens (10+ screens)
- Implemented Material 3 components & theming
- Designed Mood quadrant UI with color-coded cards
- Built Sanctuaries room list & music player UI
- Ensured accessibility & clean screen layouts

Vidney Milind Jadhav

Architecture & Stateflow

- NavHost navigation graph with typed routes
- Implemented all 7 ViewModels with StateFlow
- Integrated Socket.io for real-time chat (ChatViewModel)
- Connected Spotify Client Credentials flow
- App flow integration & feature wiring

Team Coordination Process

1. Feature planning & scope

2. Parallel development per role

3. Integration & conflict resolution

4. Manual testing on device

5. Iteration & polish

Stretch Goals — Progress



Completed



In Progress

Real-time Socket.io Chat

Fully implemented with ChatViewModel + optimistic UI

Google Maps Integration (Digital Detox)

Parks, libraries, cafes shown on live map with GPS

Spotify Mood-Based Music

Client Credentials flow + curated track suggestions by mood

Emotion Analytics (7-day insights)

Energy distribution charts + key observations + history feed

Anonymous Sanctuaries / Rooms

6 mood-aligned rooms: Healing Space, Vent Zone, Pure Joy Room

Voice journaling

Manual typing + microphone enabled journaling

Room Disclaimer Message

Rooms will display a disclaimer message to users upon entering

AI Usage & Transparency

AI tools (primarily ChatGPT) were used as a support tool during development — not as a replacement for understanding.

Where & How AI Was Used

- Brainstormed feature ideas (mood tracking, anonymous chat rooms)
- Helped debug errors in Kotlin, API calls, and Gradle issues
- Assisted in understanding MVVM, StateFlow, and API integration
- Provided guidance for UI improvements in Jetpack Compose
- Helped refine README documentation and structure

What AI Did NOT Do

- AI did not write the final code — all logic was reviewed, understood, and authored by the team
- AI did not design the app architecture — MVVM, NavGraph, and StateFlow decisions were made by the team
- AI did not replace learning — concepts were studied independently before implementation

AI Used

ChatGPT (OpenAI) — primary tool for brainstorming, debugging support, and documentation refinement

AI was a tool in our workflow — every line of code, every design decision, and every concept was owned and understood by our team.

Thank You!

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